ISE1-A2- DWM Lab Activity – 10 Marks

Title: Data Exploration, Business Intelligence Dashboard and Business Value Creation

Objective:
Analyze a real-world organizational dataset using **Power BI or Looker Studio**, identify meaningful business patterns, and propose **data-driven actions that can create value for the organization**.

Student Task

Students will work in groups and select a **real or simulated business organization** such as:

- E-commerce / Retail
- Banking
- Food Delivery
- Healthcare
- Transportation
- Education
- Manufacturing
- Hospitality
- Music
- Sports
- Transportation
- Supply chain

They must:

1. **Create Google form to collect data and collect data from ur classmates or friends**
2. Select a suitable dataset representing the organization's business operations.
3. Explore and preprocess the data.
4. Identify important business questions/problems.
5. Create meaningful visualizations.
6. Develop an **interactive Power BI or Looker Studio dashboard**.
7. Analyze the dashboard to identify trends, patterns, relationships and problems.
8. Recommend **data-driven business actions**.
9. Explain **how their analysis can create value for the organization**.
10. Present their findings through a **PowerPoint presentation**.

Business Value Component

This should be the most important part of the activity.

Students should answer:

"If you are a data analyst working for this organization, how will your analysis help the organization make better decisions and create business value?"

They should identify at least **3 business values**, for example:

Data Finding	Business Problem	Recommended Action	Business Value
Product A has declining sales	Falling demand	Run targeted promotion	Increase revenue
Region B has high customers but low sales	Low conversion	Region-specific offers	Improve sales
Product C has excessive inventory	Overstocking	Reduce future procurement	Reduce inventory cost
Customer churn is increasing	Customer loss	Retention campaign	Improve customer retention
Delivery time is high in Area X	Operational inefficiency	Optimize delivery allocation	Improve customer satisfaction

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PowerPoint Presentation

Students must prepare a **6–8 slide presentation**.

Slide 1 – Organization & Problem

- Organization/business domain
- Business problem selected
- Why the problem is important

Slide 2 – Dataset & Data Exploration

- Dataset source
- Number of records and attributes
- Important attributes
- Data quality issues

Slide 3 – Business Questions

Frame **3–5 analytical questions**, for example:

- Which product generates maximum revenue?
- Which region is underperforming?
- What is the monthly sales trend?
- Which customers are most valuable?
- Where are operational inefficiencies?

Slide 4 – Data Visualization

Show important charts and explain **why each visualization was selected**.

Slide 5 – Interactive Dashboard

Show the **Power BI/Looker Studio dashboard**.

Students should demonstrate filters/slicers and explain how the dashboard supports exploration.

Slide 6 – Key Findings

Present at least **5 insights** obtained from the analysis.

Slide 7 – Business Value & Recommendations

Clearly explain:

Finding → Business Problem → Recommendation → Expected Business Value

Examples:

- Increase revenue
- Reduce operational cost
- Improve customer retention
- Optimize inventory
- Improve resource utilization
- Improve customer satisfaction
- Identify new opportunities

Slide 8 – Conclusion

- Major findings
- Recommended actions
- Overall contribution of data analytics to the organization

Assessment Rubric – 10 Marks

Component	Marks	BL-4 Assessment
1. Data Exploration & Understanding	1.5	Analyze data quality, variables, distributions and patterns
2. Visualization & Dashboard	2.0	Select and integrate appropriate visualizations and interactive filters
3. Analytical Findings	2.0	Identify and explain meaningful trends, relationships and anomalies

Component	Marks	BL-4 Assessment
4. Business Problem Analysis	1.5	Connect data findings with actual organizational problems
5. Business Value & Recommendations	2.0	Propose actionable decisions and explain how they create value
6. PPT Presentation & Communication	1.0	Clearly communicate analysis, dashboard and recommendations
Total	**10**	

Key BL-4 Requirement

Students should **not receive full marks merely for producing a dashboard**.

They must demonstrate the chain:

Raw Data → Exploration → Visualization → Analysis → Business Problem → Recommendation → Business Value

This makes the activity strongly aligned with **BL-4 (Analyze)**.

Submission**:

- **Submit a report in PDF format that includes:**
  - **Definitions and examples of each data type.**
  - **The survey questions used for data collection.**
  - **Analysis results with appropriate visualizations.**
  - **Interpretation of the results.**
  - **Your creative project (story/comic strip or interactive visualization).**
- **Include any data files and source code used for analysis and visualization.**

Self-Learning Resources

Students independently learn either:

Microsoft Learn – Power BI Training

https://learn.microsoft.com/en-us/training/powerplatform/power-bi

or

Google Looker Studio

Youtube :

Looker Studio Tutorial (Now 'Data Studio') – Build Dashboards & Reports Step-by-Step

The existing activity already identifies Power BI and other visualization tools as resources for developing an interactive dashboard.

Final deliverables:
Dataset + Power BI/Looker Studio Dashboard + PPT Presentation + Business Value Analysis.